Personal Meeting

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1 Thresholding Derivation

1.1 Clarifications

1. λ **is real because minimization compares values.** Minimizing requires ordering objective values, and $\mathbb{C}$ has no ordering, so the objective $\frac{1}{2}\|\mathbf{x}-\mathbf{z}\|_2^2 + \lambda R(\mathbf{x})$ must be real; with $R(\mathbf{x}) \geq 0$ real, λ must be real. This is unrelated to the phase assumption $R(x) = r(|x|)$: λ would still be real for a phase-dependent penalty.

2. **The phase of $\hat{x}$ equals that of z because of a lower bound.** For fixed $|x|$ the penalty is fixed and $|x-z|^2 \geq (|x|-|z|)^2$; the bound is attained iff x has the phase of z, so the minimizer takes that phase. The polar form, the rectangular form and the triangle inequality all give this same bound and equality condition (Section 1.2, Phase).

1.2 Derivation Summary

Problem formulation.

$$\hat{\mathbf{x}} = \underset{\mathbf{x}\in\mathbb{C}^N}{\arg\min}\ \tfrac{1}{2}\|\mathbf{x}-\mathbf{z}\|_2^2 + \lambda R(\mathbf{x})$$

- $\mathbf{z} \in \mathbb{C}^N$: observed complex image.
- $\mathbf{x} \in \mathbb{C}^N$: estimate.
- $R(\mathbf{x})$: sparsity-promoting penalty, $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_0$ or $\|\mathbf{x}\|_{1/2}^{1/2}$.
- $\lambda > 0$: weight between the data term and the penalty; real, since the objective must be real-valued to be minimized.

Separability. If $R(\mathbf{x}) = \sum_i R_i(x_i)$,

$$\begin{aligned}\min_{\mathbf{x}\in\mathbb{C}^N}\ \tfrac{1}{2}\|\mathbf{x}-\mathbf{z}\|_2^2 + \lambda R(\mathbf{x}) &= \min_{\mathbf{x}\in\mathbb{C}^N}\ \sum_{i=1}^{N}\Big[\tfrac{1}{2}|x_i - z_i|^2 + \lambda R_i(x_i)\Big] \\ &= \sum_{i=1}^{N}\ \min_{x_i\in\mathbb{C}}\ \Big[\tfrac{1}{2}|x_i - z_i|^2 + \lambda R_i(x_i)\Big],\end{aligned}$$

so each $\hat{x}_i$ is found independently. Below, the index i is dropped ($x = x_i$, $z = z_i$, $R = R_i$):

$$\hat{x} = \underset{x\in\mathbb{C}}{\arg\min}\ \tfrac{1}{2}|x-z|^2 + \lambda R(x).$$

Phase. If $R(x) = r(|x|)$ (true for $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_0$, $\|\mathbf{x}\|_{1/2}^{1/2}$, defined through $|x_i|$), then for fixed $|x|$ the penalty is constant and $|x-z|^2 \geq (|x|-|z|)^2$, with equality iff x has the phase of z:

- Polar, $x = \rho e^{j\theta}$, $z = m e^{j\varphi}$: $|x-z|^2 = \rho^2 + m^2 - 2\rho m\cos(\theta-\varphi) \geq (\rho - m)^2$, "=" iff $\theta = \varphi$.
- Rectangular, $x = a + jb$, $z = c + jd$: $|x-z|^2 = |x|^2 + |z|^2 - 2(ac+bd) \geq (|x|-|z|)^2$, since $ac + bd \leq |x||z|$ (Cauchy–Schwarz), "=" iff $(a,b) = \kappa(c,d)$, $\kappa \geq 0$.
- Triangle inequality: $|x-z| \geq \big||x|-|z|\big|$, "=" iff $x = \kappa z$, $\kappa \geq 0$.

Hence, with $\rho = |x|$ and $m = |z|$,

$$g(\rho) = \tfrac{1}{2}\big(|x|-|z|\big)^2 + \lambda r(|x|) = \tfrac{1}{2}(\rho - m)^2 + \lambda r(\rho), \qquad \hat{\rho} = \underset{\rho\geq 0}{\arg\min}\ g(\rho), \qquad \hat{x} = \operatorname{sgn}(z)\,\hat{\rho}.$$

ℓ_1: soft thresholding. $r(\rho) = \rho$, $g(\rho) = \frac{1}{2}(\rho - m)^2 + \lambda\rho$, $g'(\rho) = \rho - m + \lambda$:

- $m > \lambda$: $g'(\rho) = 0 \Rightarrow \hat{\rho} = m - \lambda$.
- $m \le \lambda$: $g'(\rho) > 0$ for $\rho > 0 \Rightarrow \hat{\rho} = 0$.

$$\mathcal{S}_\lambda(z) = \operatorname{sgn}(z)\, \max(|z| - \lambda,\, 0).$$

ℓ_0: hard thresholding. $r(\rho) = \mathbf{1}[\rho > 0]$, $g(\rho) = \frac{1}{2}(\rho - m)^2 + \lambda\, \mathbf{1}[\rho > 0]$:

- $\rho = 0$: $g(0) = \frac{1}{2}m^2$.
- $\rho > 0$: best $\rho = m$, $g(m) = \lambda$.
- $\hat{\rho} = m$ iff $\lambda < \frac{1}{2}m^2$, i.e. $m > \sqrt{2\lambda}$ ($\hat{\rho} = 0$ at equality).

$$\mathcal{H}_{\sqrt{2\lambda}}(z) = z \cdot \mathbf{1}\big[|z| > \sqrt{2\lambda}\big].$$

2 $L_{1/2}$&TV Problem Splitting

Compound-regularized linear inverse problem [1], with echo $\mathbf{y} = \operatorname{vec}(\mathbf{Y})$, scene $\mathbf{h} = \operatorname{vec}(\mathbf{H})$ ($\mathbf{Y}, \mathbf{H} \in \mathbb{C}^{M\times N}$, M azimuth and N range samples; vec stacks the columns), CSA matrix $\mathbf{\Phi}$ with $\mathbf{h}_{\mathrm{CSA}} = \mathbf{\Phi}\mathbf{y}$, so the echo model is $\mathbf{\Phi}^{-1}$; total variation $\mathrm{TV}(\cdot)$ [2] and penalty weights $\lambda_1, \lambda_2 > 0$. Introducing auxiliary variables [3] and then the Lagrangian with multipliers $\xi_1, \xi_2 > 0$ (kept fixed):

$$\hat{\mathbf{h}} = \arg\min_{\mathbf{h}} \ \big\|\mathbf{y} - \mathbf{\Phi}^{-1}\mathbf{h}\big\|_2^2 + \lambda_1\|\mathbf{h}\|_{1/2} + \lambda_2\,\mathrm{TV}(\mathbf{h}) \tag{1}$$

$$= \arg\min_{\mathbf{h},\mathbf{z}_1,\mathbf{z}_2} \ \big\|\mathbf{y} - \mathbf{\Phi}^{-1}\mathbf{h}\big\|_2^2 + \lambda_1\|\mathbf{z}_1\|_{1/2} + \lambda_2\,\mathrm{TV}(\mathbf{z}_2) \quad \text{s.t.} \ \ \|\mathbf{h} - \mathbf{z}_1\|_2^2 = 0, \ \ \|\mathbf{h} - \mathbf{z}_2\|_2^2 = 0 \tag{2}$$

(introduce auxiliary variables $\mathbf{z}_1, \mathbf{z}_2$: $\|\mathbf{h} - \mathbf{z}\|_2^2 = 0 \iff \mathbf{z} = \mathbf{h}$)

$$\approx \arg\min_{\mathbf{h},\mathbf{z}_1,\mathbf{z}_2} \ \underbrace{\big\|\mathbf{y} - \mathbf{\Phi}^{-1}\mathbf{h}\big\|_2^2 + \lambda_1\|\mathbf{z}_1\|_{1/2} + \lambda_2\,\mathrm{TV}(\mathbf{z}_2) + \xi_1\|\mathbf{h} - \mathbf{z}_1\|_2^2 + \xi_2\|\mathbf{h} - \mathbf{z}_2\|_2^2}_{\Gamma(\mathbf{h},\mathbf{z}_1,\mathbf{z}_2)} \tag{3}$$

(Lagrangian: move the constraints into the objective)

The minimization over $(\mathbf{h}, \mathbf{z}_1, \mathbf{z}_2)$ returns its $\mathbf{h}$-component. With ξ_1, ξ_2 fixed, $\xi\|\mathbf{h} - \mathbf{z}\|_2^2$ penalizes a copy that drifts away from $\mathbf{h}$ [3], so (3) approximates (2).

Alternating minimization of Γ (one variable at a time, the others at their latest values; terms without that variable are constants) splits (3) into three subproblems:

$$\mathbf{h}^{(k+1)} = \arg\min_{\mathbf{h}} \ \Gamma\big(\mathbf{h}, \mathbf{z}_1^{(k)}, \mathbf{z}_2^{(k)}\big) = \arg\min_{\mathbf{h}} \ \big\|\mathbf{y} - \mathbf{\Phi}^{-1}\mathbf{h}\big\|_2^2 + \xi_1\big\|\mathbf{h} - \mathbf{z}_1^{(k)}\big\|_2^2 + \xi_2\big\|\mathbf{h} - \mathbf{z}_2^{(k)}\big\|_2^2, \tag{4}$$

$$\mathbf{z}_1^{(k+1)} = \arg\min_{\mathbf{z}_1} \ \Gamma\big(\mathbf{h}^{(k+1)}, \mathbf{z}_1, \mathbf{z}_2^{(k)}\big) = \arg\min_{\mathbf{z}_1} \ \xi_1\big\|\mathbf{z}_1 - \mathbf{h}^{(k+1)}\big\|_2^2 + \lambda_1\|\mathbf{z}_1\|_{1/2}, \tag{5}$$

$$\mathbf{z}_2^{(k+1)} = \arg\min_{\mathbf{z}_2} \ \Gamma\big(\mathbf{h}^{(k+1)}, \mathbf{z}_1^{(k+1)}, \mathbf{z}_2\big) = \arg\min_{\mathbf{z}_2} \ \xi_2\big\|\mathbf{z}_2 - \mathbf{h}^{(k+1)}\big\|_2^2 + \lambda_2\,\mathrm{TV}(\mathbf{z}_2). \tag{6}$$

- (4): quadratic in $\mathbf{h}$, solved in closed form below.
- (5): $L_{1/2}$ denoising of $\mathbf{h}^{(k+1)}$, solved entrywise by half thresholding [4].
- (6): TV denoising of $\mathbf{h}^{(k+1)}$, solved by Chambolle's projection algorithm [5].

In practice $\xi_1 = \xi_2 = \xi$. Dividing by ξ, (5) minimizes $\|\mathbf{z}_1 - \mathbf{h}^{(k+1)}\|_2^2 + (\lambda_1/\xi)\|\mathbf{z}_1\|_{1/2}$ and (6) minimizes $\frac{1}{2}\|\mathbf{z}_2 - \mathbf{h}^{(k+1)}\|_2^2 + \big(\lambda_2/(2\xi)\big)\,\mathrm{TV}(\mathbf{z}_2)$, so the effective weights λ_1/ξ and $\lambda_2/(2\xi)$ are tuned through λ_1, λ_2.

2.1 References

[1] M. Liu, J. Pan, J. Zhu, Z. Chen, B. Zhang, and Y. Wu, "A sparse SAR imaging method for low-oversampled staggered mode via compound regularization," *Remote Sens.*, vol. 16, no. 8, Art. no. 1459, 2024.

[2] L. I. Rudin, S. Osher, and E. Fatemi, "Nonlinear total variation based noise removal algorithms," *Physica D*, vol. 60, no. 1–4, pp. 259–268, 1992.

[3] Y. Wang, J. Yang, W. Yin, and Y. Zhang, "A new alternating minimization algorithm for total variation image reconstruction," *SIAM J. Imaging Sci.*, vol. 1, no. 3, pp. 248–272, 2008.

[4] Z. Xu, X. Chang, F. Xu, and H. Zhang, "$L_{1/2}$ regularization: A thresholding representation theory and a fast solver," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 23, no. 7, pp. 1013–1027, 2012.

[5] A. Chambolle, "An algorithm for total variation minimization and applications," *J. Math. Imaging Vis.*, vol. 20, pp. 89–97, 2004.

3 Subproblem Derivation

3.1 Subproblem 1: Image Update

$\mathbf{\Phi}$ is unitary ($\mathbf{\Phi}^{-1} = \mathbf{\Phi}^H$), so $\|\mathbf{y} - \mathbf{\Phi}^{-1}\mathbf{h}\|_2^2 = \|\mathbf{\Phi}^H(\mathbf{h}_{\mathrm{CSA}} - \mathbf{h})\|_2^2 = \|\mathbf{h}_{\mathrm{CSA}} - \mathbf{h}\|_2^2$. Writing each squared norm as a sum over the MN entries, with $[v_n]_{n=1}^{MN}$ the vector with entries v_n:

$$
\begin{aligned}
\mathbf{h}^{(k+1)} &= \arg\min_{\mathbf{h}} \left\|\mathbf{y} - \mathbf{\Phi}^{-1}\mathbf{h}\right\|_2^2 + \xi_1\left\|\mathbf{h} - \mathbf{z}_1^{(k)}\right\|_2^2 + \xi_2\left\|\mathbf{h} - \mathbf{z}_2^{(k)}\right\|_2^2 \\
&= \arg\min_{\mathbf{h}} \left\|\mathbf{h}_{\mathrm{CSA}} - \mathbf{h}\right\|_2^2 + \xi_1\left\|\mathbf{h} - \mathbf{z}_1^{(k)}\right\|_2^2 + \xi_2\left\|\mathbf{h} - \mathbf{z}_2^{(k)}\right\|_2^2 \\
&\qquad (\mathbf{\Phi} \text{ unitary}) \\
&= \arg\min_{\mathbf{h}} \sum_{n=1}^{MN}\Big[\left|h_{\mathrm{CSA},n} - h_n\right|^2 + \xi_1\left|h_n - z_{1,n}^{(k)}\right|^2 + \xi_2\left|h_n - z_{2,n}^{(k)}\right|^2\Big] \\
&\qquad (\|\mathbf{v}\|_2^2 = \sum_n |v_n|^2) \\
&= \arg\min_{\mathbf{h}} \sum_{n=1}^{MN}\Big[\big(\mathrm{Re}\{h_n\} - \mathrm{Re}\{h_{\mathrm{CSA},n}\}\big)^2 + \big(\mathrm{Im}\{h_n\} - \mathrm{Im}\{h_{\mathrm{CSA},n}\}\big)^2 \\
&\qquad\qquad + \xi_1\big(\mathrm{Re}\{h_n\} - \mathrm{Re}\{z_{1,n}^{(k)}\}\big)^2 + \xi_1\big(\mathrm{Im}\{h_n\} - \mathrm{Im}\{z_{1,n}^{(k)}\}\big)^2 \\
&\qquad\qquad + \xi_2\big(\mathrm{Re}\{h_n\} - \mathrm{Re}\{z_{2,n}^{(k)}\}\big)^2 + \xi_2\big(\mathrm{Im}\{h_n\} - \mathrm{Im}\{z_{2,n}^{(k)}\}\big)^2\Big] \\
&\qquad (|u - h_n|^2 = (\mathrm{Re}\{h_n\} - \mathrm{Re}\{u\})^2 + (\mathrm{Im}\{h_n\} - \mathrm{Im}\{u\})^2) \\
&= \frac{1}{1+\xi_1+\xi_2}\Big[\mathrm{Re}\big\{h_{\mathrm{CSA},n} + \xi_1 z_{1,n}^{(k)} + \xi_2 z_{2,n}^{(k)}\big\} + j\,\mathrm{Im}\big\{h_{\mathrm{CSA},n} + \xi_1 z_{1,n}^{(k)} + \xi_2 z_{2,n}^{(k)}\big\}\Big]_{n=1}^{MN} \\
&\qquad \text{(the sum separates into } 2MN \text{ one-dimensional quadratics, one in each } \mathrm{Re}\{h_n\} \text{ and } \mathrm{Im}\{h_n\}; \\
&\qquad\ \text{minimize each by setting its derivative to zero)} \\
&= \frac{\mathbf{h}_{\mathrm{CSA}} + \xi_1\mathbf{z}_1^{(k)} + \xi_2\mathbf{z}_2^{(k)}}{1+\xi_1+\xi_2}. \qquad (7) \\
&\qquad (\mathrm{Re}\{c\} + j\,\mathrm{Im}\{c\} = c \text{ for each entry})
\end{aligned}
$$